

Abel Sabu

Portfolio | E-mail | LinkedIn | Github

SUMMARY

Computer Science student specializing in **AI/ML** and **Cybersecurity** with expertise in Python, reinforcement learning, and cloud computing using **Azure** services, **Docker**, and **CI/CD** pipelines. Experienced in full-stack development with **React**, **Node.js**, and microservices architecture, combined with hands-on IT infrastructure management including Active Directory and **SCCM** deployment. Proven ability to deliver scalable **MLOps** solutions and agile development practices in enterprise environments.

EDUCATION

York University

*Bachelor in Computer Science with Specialization in **Computer Security** and **AI***

Toronto, ON

Sep. 2021 – Dec. 2026

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, Shell - Scripting

Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI

Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, Microsoft Office, Power BI

Libraries: pandas, NumPy, Matplotlib

Awards: Won "**Rookie Of The Year**" Award for Client Services, Won Top 8 ideas for UNHACK 2023

Relevant Courses: Calculus, Net-Centric Web Development/Mobile Computing, Object-oriented programming, Data Structures & Algorithms, Computer Network and Protocols, Malware Analysis, Applied Cryptography, Full-Stack, Operating Systems, TCP/IP - Protocols/layers

Summits: Attended "AWS SUMMIT 2025" to learn more about the industry and gained a lot of meaningful connections

EXPERIENCE

Information Technology Assistant

May 2024 – Sep. 2024

York University

Toronto, ON

- Reduced average ticket resolution time by 10% by collaborating with coordinator and cross-functional team to streamline ticketing system workflows and implemented process improvements.
- Accelerated PC deployment setup by 0.5 hours/devices by managing device imaging for 100+ new PCs and laptops using SCCM while maintaining security compliance standards.
- Enhanced software deployment efficiency by 46% by developing configuration management policies supporting SCCM cloud migration initiatives.

Client Service Representative - Team Lead

Sept. 2023 – Present

York University

Toronto, ON

- Improved client satisfaction scores by 30% by managing almost 1,000 daily visitor interactions and memberships at Sport and Recreation facilities by reducing invoice processing time by 10% and increased staff training efficiency by 12% by training and mentoring staff on membership processes, facility rentals, and client-service best practices.
- Delivered exceptional client service by handling inquiries, processing renewals, and coordinating with high-profile renting partners like the *Toronto Raptors*, *Team Canada (Volleyball and Basketball)*, *National Bank of Canada*.
- Partnered** with the Sport and Recreation coordinator and manager to **refine** and enhance team performance by almost 50% by leading regular huddles and workshops to implement RATER customer-service model and monitor key performance metrics.

PROJECTS

Precogni – Driver Drowsiness Detector (Hack the 6ix 2025) | Python, Gemini API, Node.js, OpenCV, Whisper AI

- Tackled the safety gap of reactive fatigue alerts by prototyping a proactive driver "**sleep preventer**" during a 3-day hackathon.
- Integrated the **Gemini AI API** into a computer-vision pipeline to read early fatigue cues (eye closure, yawning, head tilt) and trigger preventive engagement.
- Built the alerting loop, tuned thresholds, and coordinated **CV + AI integration** across a 4-person team under tight time constraints.

SaveBetter – Adaptive Financial Planning App (Kiro [AWS] Hackathon) | *Next.js, React 18, TypeScript, Tailwind*

- Developed adaptive saving engine with sophisticated algorithms and **AI-enhanced** behavioral **analysis** using **OpenAI GPT-4** for personalized financial recommendations and bias detection and a dual assessment system with **machine learning** pattern recognition, confidence scoring, and real-time Canadian banking data integration from major institutions **RBC, TD, BMO**.
- Engineered a production-ready framework for GPT-4 integration—env-keyed switch, typed interfaces, and prompts—enabling seamless upgrade from simulated insights to real AI analysis.

RetailAI: Microsoft Azure Hackathon | *Azure ML, Custom Vision, Azure OpenAI, Node.js, Kubernetes*

- Built retail AI platform using **Azure ML, Custom Vision** for image-based product search, and RAG-powered **chatbot** with Azure **OpenAI** for natural language inventory queries.
- Developed data **pipelines** for anomaly detection, integrated structured/unstructured data via **Cognitive Search**, and deployed **full-stack** ML solutions with **Node.js** and **Azure** services.
- Architected **MLOps** framework with Azure **DevOps**, **MLflow**, and Infrastructure as Code for one-click model deployments, automated drift detection, and event-driven microservices on Azure **Kubernetes** Service with **Cosmos DB** monitoring.